

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th July 2022

Subject SP4 – Pensions and Other Benefits

Time allowed: 3 Hours 30 Minutes (14.30 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. *Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.***
- 2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.***
- 3. *Mark allocations are shown in brackets.***

- Q .1)** A BSE100 listed company sponsors a hybrid pension plan for a certain category of employees. The plan has characteristics of both a Defined Benefit Plan and a Defined Contribution Plan. The plan is treated as a defined benefit plan for accounting purposes. The benefit formula is expressed as a notional account balance, with employer service credits provided and interest credits earned on the account balance.

The employer service credit is a percentage of salary, which varies by age and service. The interest credit is a specific rate; and is tied to the yield on Central Government bonds. The interest credit rate can be separate from the actual returns earned by the pension trust. Members may choose to take a lump sum or an annuity at termination or retirement. Annuities are paid out directly by the fund.

- i)** Describe the risks of the above hybrid plan from the perspective of the company. (5)
- ii)** Describe the ways the company could manage the risks associated with this plan. (5)

The CFO of the company is considering terminating this plan and replacing it with one of the following options:

- a) Defined contribution pension plan
 - b) Variable Defined Benefit pension plan
 - c) One-time salary increase
- iii)** Describe the advantages and disadvantages of each option from the perspectives of both the employee and the employer. (10)
- [20]**

- Q. 2)** You are the Actuary for a reputed company that is known for its paternalistic policies towards its employees. The company sponsors a post-retirement medical plan which is still open to new entrants. The plan is unfunded and the benefits are provided on a pay-as-you-go basis.

- i)** List the key assumptions used in the valuation of a Post-Retirement Medical Plan. (4)
- ii)** Explain why prevalence of post-retirement medical plans may be reducing and why plan sponsors may be looking to reduce such benefits. (6)

The Company also sponsors a tax approved Defined Benefit Pension Plan that is underfunded. Due to a recent divestiture, the Company has extra cash and is trying to decide the best use for this money.

The Company is considering the following options:

Option 1: Make a large lump sum contribution to the tax approved pension plan

Option 2: Pay a one-time bonus that would be immediately taxable to the employees

Option 3: Fund the Post-Retirement Medical Plan

iii) Discuss the advantages and disadvantages of each of these options from the perspectives of the following stakeholders:

- a) Active employees
- b) Shareholders
- c) Company.

(20)
[30]

Q. 3) A large benefit scheme includes pension benefits that are linked to the final salaries of members.

The assets of the scheme are invested in a range of equities, fixed interest and property investments without any formal regard to the matching of the assets to the liabilities.

You are the scheme Actuary. In order to deal with the regulatory provisions, the Finance Controller suggests purchasing only those assets whose value and timings meet the value and timing of the liability.

i) Discuss the issues the scheme would need to consider. (6)

ii) Outline the difficulties the scheme might experience in trying to implement the same. (6)

Given the difficulties in managing the DB scheme, the employer has recently decided to close the scheme to new members and cease the accrual of future service benefits for existing members. The employer is thinking of a new arrangement for new members.

iii) Outline the regulatory provisions that may have influenced the decision to close the scheme. (10)

iv) Provide the options available for the provisions of the outstanding benefit payments to the existing members. (3)

The Trustees have decided that the scheme will now operate as an independent entity without any obligation to the employer. The Trustees are concerned about running the scheme and have decided to earmark expenses for better management of scheme.

v) Discuss the factors the Trustees should have considered for earmarking the future expenses. (12)

Three years later, the Trustees have reviewed the expenses that have been incurred during and since the closure of the scheme. This review has concluded that, over the period, expenses for investment management and professional services were significantly higher than assumed.

vi) Suggest, with reasons, why this overspend may have arisen. (8)

The trustees are now considering offering the option of the transfer of funds to the beneficiaries to extinguish the liabilities as they believe that it is the safest option to have the benefits in hand given the current situation.

vii) Outline the issues you believe the trustees should consider before adopting this policy. (5)
[50]
